

1 Critical evaluation of the modified Gompertz response
2 function as a parametric template for
3 treatment-response trajectories in N-of-1 trial
4 simulation

5 pmsimstats team

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35 1 Abstract

36 **Background.** Simulation-based power analyses for aggregated N-of-1 trials re-
37 quire a parametric template for the within- participant response trajectory. The
38 pmsimstats framework adopts a modified Gompertz function for each of three
39 latent response components (biological response, placebo-belief response, time-
40 variant natural-history component). The choice of family is loaded into the simu-
41 lated data before any analysis-model question is asked, so the implications of the
42 family for power, Type I error, and bias are foundational rather than peripheral.

43 **Methods.** We evaluate the modified Gompertz family against the relevant biolog-
44 ical, pharmacokinetic, and biostatistical literature. We characterize the parameter
45 space the function covers, the trajectories it can and cannot represent, and the
46 sensitivity of standard pmsimstats inference to misspecification of the response-
47 curve family. Three alternative families are compared as sensitivity analyses:
48 the logistic (symmetric sigmoidal), the hyperbolic-tangent (fast-saturating), and
49 a piecewise-linear breakpoint model (no smoothness assumption). For each al-
50 ternative we compute power, Type I error, and bias of the biomarker-treatment
51 interaction estimate at trial-relevant sample sizes against simulated data drawn
52 from the alternative family but analyzed under the standard linear-mixed model.

53 **Results.** A $1\{, \}$ 000-replicate-per-cell simulation across 16 cells (4 trajectory fam-
54 ilies $\times$ 2 data-generating process (DGP) architectures $\times$ 2 biomarker effect-size
55 levels) at $N = 35$, open-label plus blinded discontinuation (OL+BDC) design,
56 $t_{1/2} = 0$, found that the inferential consequence of trajectory family is architecture-
57 dependent. Under Architecture B (multivariate normal (MVN) differential corre-
58 lation), where the biomarker-treatment interaction lives in the covariance struc-
59 ture and is therefore mean-independent, trajectory family is exactly inert: with
60 matched random-number seeds the four families return identical rejection deci-
61 sions, so the production-seed spread $\{0.774, 0.777, 0.791, 0.804\}$ is sampling noise.
62 Under Architecture A (direct mean moderation), implemented in its faithful mul-
63 tiplicative form in which the biomarker moderation rides the response trajectory,
64 the four families separate: powers were $\{0.743, 0.760, 0.778, 0.782\}$ for Gompertz,
65 hyperbolic-tangent, piecewise-linear, and logistic respectively, a 0.039 spread in
66 which the Gompertz default sits roughly two Monte Carlo standard errors (MCSE)
67 below the logistic and is the lowest-power, that is mildly conservative, family.
68 Type I error at the null sat in $[0.019, 0.038]$ across all eight null cells, slightly

69 conservative and without family-specific inflation; bias of the interaction estimate
70 was negligible and family-invariant (≈ -0.23).

71 **Conclusions.** The modified Gompertz family is a defensible default for the
72 response-curve template in pmsimstats N-of-1 simulation, with two qualifications.
73 Under a covariance-channel (MVN) data-generating process, trajectory family can-
74 not affect the interaction test by construction, so the choice is inferentially free.
75 Under a mean-moderation data-generating process, trajectory family does carry
76 a modest inferential cost at the few-percent level, and the Gompertz default is
77 mildly conservative relative to the symmetric and breakpoint alternatives. We rec-
78 ommend reporting Gompertz-vs-alternative sensitivity at least once per parameter
79 regime where simulation results inform protocol-design decisions, with particular
80 attention to mean-moderation regimes, and we propose a default sensitivity grid
81 that the framework can run alongside its primary cells.

82 2 Introduction

83 2.1 A claim about response-curve choice in N-of-1 trial simulation

84 [1]

85 **The Gompertz function is a defensible default for the**
86 **saturating-shape regime in N-of-1 simulation.**

- 87 • Its asymmetric three-parameter form spans profiles that logistic
88 and exponential families cannot represent.
- 89 • Established precedent in tumour dynamics, microbial growth,
90 and the unified-Richards family supports the choice.
- 91 • The claim is conditional on the saturating-shape regime charac-
92 teristic of most pharmacodynamic trajectories over typical trial
93 timescales.

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96 The first claim is that the modified Gompertz function used as the
97 parametric template for treatment-response trajectories in aggregated
98 N-of-1 trial simulation (specifically, in the pmsimstats simulation
99 framework that operationalizes the Hendrickson et al. design family¹)
100 is a defensible default for the saturating-shape regime that character-
101 izes most pharmacodynamic and chronic-condition response curves
102 over the typical trial timescale. The function is well-established in
103 tumour growth dynamics^{2,3}, microbial growth modeling⁴, and the
104 broader unified- Richards growth-curve family⁵⁻⁷; its three-parameter
105 form is flexible enough to capture the asymmetric saturation profiles
106 that linear extrapolation, exponential decay, or symmetric logistic
107 models cannot.

108 [2]

Family-misspecification sensitivity is absent from the N-of-1 literature and should be reported routinely.

- Existing N-of-1 simulation studies inherit trajectory families without comparing alternatives.
- Oncology and pharmacology modeling have established comparative apparatus that the N-of-1 field has not imported.
- Routine sensitivity reporting is needed rather than silent assumption of family adequacy.

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The second claim is that family-misspecification sensitivity is essentially absent from the published N-of-1 methodology literature, and that it should be reported routinely rather than assumed away. The methodological N-of-1 corpus^{1,8–10} is, to the best of our knowledge, silent on parametric trajectory shape: papers either inherit a chosen family from the upstream literature without comparing alternatives, or specify a linear or AR(1) form that is too coarse to capture the saturation that real trajectories exhibit. The oncology-modeling literature has set the precedent for what a proper family-comparison sensitivity analysis looks like^{6,11}, and the pharmacology literature has converged on Hill / sigmoid-Emax / indirect-response models for related questions^{12,13}. The N-of-1 literature has not imported this comparative apparatus, and the present paper endeavors to supply it.

[3]

The case for Gompertz is incomplete without a direct comparison of inferential consequences across alternative families.

- The paper does not assert Gompertz is uniquely correct but argues it is reasonable within the unified-Richards class.
- Inferential consequences (power, Type I error, bias) under alternative families must be quantified, not asserted.
- The simulation study uses the standard pmsimstats linear-mixed analysis at trial-relevant sample sizes.

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The two claims are coupled because the case for a chosen family is incomplete without the comparison. We do not argue that Gompertz is the correct family in any strict sense; rather, we argue that it is a reasonable choice within the unified-Richards family of saturating curves, and that the inferential consequences of choosing it (versus the symmetric logistic, the von Bertalanffy growth curve¹⁴, or a piecewise-linear breakpoint model) should be quantified rather than asserted.

151 The paper reports a simulation study that does so for the standard
152 pmsimstats linear-mixed analysis at trial-relevant sample sizes.

153 2.2 Why the question matters

154 [4]

155 **Family choice is a DGP-level decision whose inferential cost**
156 **cannot be detected without varying the DGP family while**
157 **holding the analysis fixed.**

- 158 • A single-family simulation cannot reveal the inferential cost of
159 that family choice.
- 160 • Robustness to family is a joint property of the DGP and analysis
161 specification.
- 162 • This is analysis-model misspecification logic applied one level
163 down in the simulation hierarchy.

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166 First, the chosen family loads into the data before any analysis-side
167 question is asked. A power simulation that draws every trajectory
168 from a Gompertz curve and then analyses each under a standard lin-
169 ear mixed-effects model with treatment indicator and AR(1) residual
170 correlation cannot detect the inferential cost of the family choice it-
171 self. Robustness of the inference to the family is a property of the
172 joint DGP-and-analysis specification; only a comparative simulation
173 that varies the DGP family while holding the analysis fixed can quan-
174 tify it. This is the same logic that underpins analysis-model misspec-
175 ification studies in the broader clinical-trial methodology literature¹⁵,
176 applied one level down in the simulation hierarchy, and it appears to
177 us that the N-of-1 simulation literature has not yet drawn the parallel
178 explicitly.

179 [5]

180 **The Gompertz parameterization has known identifiabil-**
181 **ity fragility in short noisy trajectories that has not been**
182 **characterized for N-of-1 designs.**

- 183 • Asymptote and rate parameters correlate strongly; the shape pa-
184 rameter is near-unidentifiable with few inflection- region time-
185 points.
- 186 • Unified-Richards reparameterizations and sloppiness diagnostics
187 exist but have not been applied to N-of-1 trajectory data.
- 188 • Comparative identifiability results from tumour-growth modeling
189 support the need for an analogous N-of-1 diagnosis.

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Second, the function family has known identifiability fragility in short noisy trajectories. Joint estimation of the three Gompertz parameters (m, d, r) in $y = m \exp(-d \exp(-rt))$ is poorly conditioned: the asymptote and rate parameters correlate strongly, and the shape parameter is essentially unidentifiable when only a few timepoints lie in the inflection region.⁶ and⁷ propose unified-Richards reparameterizations in which the parameters become biologically interpretable (maximum relative growth rate, age at inflection, value at inflection, asymptote) with lower correlation, and¹⁶ develops the formal ‘sloppiness’ diagnostic that quantifies this directly as a property of the parameter space.¹¹ fitted eight tumour-growth models (exponential, exponential-linear, power-law, Gompertz, logistic, generalized logistic, von Bertalanffy, dynamic-carrying-capacity) to two in vivo systems and ranked them on goodness-of-fit, parametric identifiability, and predictive power; the power-law model was the most identifiable in lung data while Gompertz was best for breast. The same kind of comparative diagnosis has not been performed for short N-of-1 trajectories, and its absence is, in our view, problematic.

[6]

The Gompertz default in N-of-1 simulation reflects historical borrowing from growth biology, not empirical comparison with pharmacology alternatives.

- The pmsimstats Gompertz form traces to tumour dynamics and microbial growth literature, not pharmacodynamic evidence.
- The pharmacology literature defaults to Hill/sigmoid-Emax and indirect-response models for comparable questions.
- The two traditions have not been compared; inheritance is not itself evidence of adequacy.

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Third, the default in N-of-1 simulation reflects historical borrowing rather than empirical optimality. The Gompertz form entered the pmsimstats framework via the broader growth-curve tradition, where its use traces to Laird’s tumour dynamics work² and to Zwietering’s microbial growth reformulations⁴. The pharmacology literature, addressing related questions about dose-response and time-course of pharmacological action, defaults to the Hill / sigmoid-Emax family^{12,17} and to Jusko’s indirect-response models¹³. These two traditions have largely not spoken to each other; the N-of-1 simulation literature has inherited the growth-biology default without an explicit defense against the pharmacology default. Either default may be adequate for the satura-

tion regime, but the inheritance is not itself evidence, and the question merits a direct empirical answer.

2.3 What the published N-of-1 literature does and does not do

[7]

No published N-of-1 simulation study has examined parametric trajectory-family choice or reported family-misspecification sensitivity.

- CONSORT-N-of-1, foundational programmatic statements, and the Hendrickson framework all omit trajectory-family comparison.
- Silent inheritance of the Gompertz form is pervasive across applied N-of-1 simulation work.
- Adjacent literatures (oncology, pharmacometrics, latent-class growth) have comparative apparatus that has not been applied to N-of-1 simulation.

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A review of the recent N-of-1 methodology literature confirms, to the best of our knowledge, a silence on the parametric-trajectory-family question.

[8]

Each major N-of-1 methodological strand silently inherits the Gompertz default without family comparison.

- CONSORT-N-of-1 reporting standards, Lillie and Schork programmatic statements, and the Hendrickson framework all omit family comparison.
- Applied prazosin-PTSD work inherits the framework without revisiting the trajectory assumption.
- No published N-of-1 simulation study, to the authors' knowledge, has broached the misspecification question.

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The CONSORT extension for N-of-1 trials¹⁰ codifies reporting standards but does not address parametric trajectory shape. The foundational programmatic statements^{8,9} emphasize design and analysis broadly without committing to a parametric response form. The Hendrickson et al. framework¹ introduces the modified Gompertz template into the simulation engine for biomarker-validation power analyses without comparing it against alternatives. Recent applied papers

in chronic-condition N-of-1 trials (including the prazosin-PTSD reference work building on¹⁸) inherit the framework without examining the choice. Unfortunately, this pattern of silent inheritance is pervasive enough that no published N-of-1 simulation study, to the best of our knowledge, has broached the family-misspecification question.

[9]

Comparative trajectory-family apparatus exists in adjacent literatures but has not been applied to N-of-1 simulation.

- Oncology modeling, pharmacometrics, and latent-class growth literature each have established multi-family comparison methods.
- None of these methods has been applied with the Gompertz form as the N-of-1 simulation reference.
- Importing this machinery is the explicit contribution of the present paper.

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In adjacent literatures the comparative apparatus does exist. The oncology-modeling community routinely fits multiple growth families and compares them on identifiability and predictive performance¹¹. The pharmacometrics community produces explicit sensitivity studies for sigmoid-Emax and indirect-response models^{19,20}. The latent-class growth-trajectory literature treats response-curve heterogeneity through mixture modeling²¹. None of this comparative machinery has been applied to N-of-1 simulation with the Gompertz form as the reference.

[10]

The paper’s contribution is a four-family comparison against the standard pmsimstats analysis, reported under the ADEMP framework.

- Four families are compared: Gompertz, symmetric logistic, hyperbolic-tangent, and piecewise-linear breakpoint.
- The analysis model is held fixed at the standard pmsimstats linear-mixed specification.
- Results are reported in the Morris-White-Crowther ADEMP framework adopted as the project-wide standard.

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The present paper’s contribution is to import that machinery, specifically: a four-family comparison (Gompertz, symmetric logistic, hyperbolic-tangent, piecewise-linear breakpoint) crossed against the standard pmsimstats linear-mixed analysis, with results reported in

the Morris-White-Crowther¹⁵ ADEMP framework already adopted as
the project-wide simulation-reporting standard.

2.4 What this paper contributes

[11]

Section 2 characterizes the pmsimstats Gompertz parameterization and its relationship to the unified-Richards family.

- The vertical-offset adjustment and calibration parameter sets are documented.
- The unified-Richards reparameterization provides more interpretable, less correlated parameters.
- This section establishes the reference point for all subsequent sensitivity comparisons.

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In section 2 we characterize the modified Gompertz family as implemented in pmsimstats: the vertical-offset adjustment, the parameter regions corresponding to the calibration sets used across published pmsimstats simulation studies, and the unified-Richards reparameterization^{6,7} that supplies more interpretable and less correlated parameters.

[12]

Section 3 surveys three alternative families at matched marginal effect size for visual and structural comparison with Gompertz.

- Families covered: symmetric logistic, hyperbolic-tangent, piecewise-linear breakpoint.
- Each is calibrated to match the Gompertz reference at the same asymptote and anchor-point response level.
- Shape differences (symmetry, curvature, smoothness) are the primary axes of contrast.

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In section 3 we survey three alternative response-curve families: the symmetric logistic²², the hyperbolic-tangent (formally a rescaled symmetric logistic), and a piecewise-linear breakpoint model. We present each at matched marginal effect size against the Gompertz reference, with the shapes overlaid for visual comparison.

[13]

351 **Sections 4 and 5 report a Monte Carlo sensitivity simula-**
352 **tion comparing power, Type I error, and bias across the four**
353 **trajectory families.**

- 354 • Each family generates trajectories; all are analyzed under the
355 fixed standard pmsimstats linear-mixed model.
- 356 • Performance measures are reported with conventional MCSEs.
- 357 • The design borrows the Benzekry eight-model comparative
358 methodology and the Jagadeesan sloppiness diagnostic.

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361 In sections 4 and 5 we conduct a Monte Carlo sensitivity-analysis
362 simulation that generates trajectories under each of the four fami-
363 lies and analyses them under the standard pmsimstats linear-mixed
364 model. We report power, Type I error, and bias of the biomarker-
365 treatment interaction estimate under each cell, with conventional MC-
366 SEs. The simulation borrows the comparative methodology of¹¹ (eight-
367 model tumour-growth comparison) and the sloppiness diagnostic of¹⁶
368 for small-sample identifiability assessment.

369 [14]

370 **Section 6 identifies the trajectory regimes where Gompertz**
371 **misspecification would be inferentially consequential and rec-**
372 **ommends sensitivity-grid extensions.**

- 373 • Consequential regimes include non-saturating growth, biphasic
374 patterns, and breakpoint behavior.
- 375 • Standard N-of-1 inference is materially affected by family choice
376 in these regimes.
- 377 • Specific sensitivity-grid extensions are recommended for pmsim-
378 stats users.

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381 In section 6 we identify the trajectory regimes in which Gompertz mis-
382 specification would be expected to be consequential: non-saturating
383 growth beyond the trial window, biphasic patterns, and breakpoint be-
384 havior. For these regimes, the standard N-of-1 inference is materially
385 affected by the family choice, and we recommend specific sensitivity-
386 grid extensions that pmsimstats users should run alongside their pri-
387 mary cells.

388 [15]

389 **A companion technical document preserves full mathematical**
390 **and historical context omitted from the manuscript proper.**

- Location: docs/25-gompertz-model-evaluation.md.
- Contains citations, derivations, and historical context that did not survive manuscript editing.
- The manuscript is self-contained; the companion document is supplementary reference material.

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The full mathematical and historical context, including citations that did not survive editing into the manuscript proper, is preserved in the companion technical document at docs/25-gompertz-model-evaluation.md.

3 The modified Gompertz function in pmsimstats

[16]

The pmsimstats framework adopts a modified monotone-saturating Gompertz function for each of three latent response components.

- Provenance traces to Gompertz’s 1825 mortality law and Winsor’s 1932 growth-curve application.
- Three components are parameterized: biological response BR, placebo-belief response PB, and natural-history TV.
- The modified form is monotone-saturating, distinguishing it from the original mortality-curve application.

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The Gompertz curve originates in Gompertz’s 1825 law of human mortality²³ and was first applied as a general growth curve by²⁴; the pmsimstats framework adopts the modified, monotone-saturating form of that curve. The framework parameterizes each of the three latent response components (biological response BR, placebo-belief response PB, and time-variant natural-history component TV) with a modified Gompertz function

$$y(t) = \text{maxr} \cdot \exp(-\text{disp} \cdot \exp(-\text{rate} \cdot t)), \quad (1)$$

[17]

The three-parameter Gompertz form is asymmetric and monotone-saturating, with the inflection location determined jointly by rate and displacement.

- Parameters are maxr (asymptote), rate (approach speed), and disp (curvature near $t = 0$); a vertical offset ensures $y(0) = 0$.

- Inflection at $t^* = \log(\text{disp})/\text{rate}$; the asymmetry is the key structural distinction from logistic.
- Prazosin/PTSD reference values are $\text{maxr}_{BR} \approx 4$, $\text{rate}_{BR} \approx 0.5$, $\text{disp}_{BR} \approx 4$.

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with vertical-offset adjustment so that $y(0) = 0$. The three parameters are maxr (asymptotic maximum response, matching the on-drug or on-natural-history saturation level), rate (which controls how quickly the response approaches its asymptote), and disp (which controls the curvature near $t = 0$). The Gompertz form is monotone-saturating and asymmetric: the inflection occurs at $t^* = \log(\text{disp})/\text{rate}$, with the rise being relatively slower before the inflection and faster after. The parameterization used at the prazosin/PTSD reference cell sets $\text{maxr}_{BR} \approx 4$ nightmares per week, $\text{rate}_{BR} \approx 0.5$ per week, and $\text{disp}_{BR} \approx 4$ for biological response; analogous parameter sets for PB and TV are documented in docs/25-gompertz-model-evaluation.md and R/utilities.R.

4 Alternative response-curve families

[18]

Three alternative families are calibrated to match Gompertz at a common anchor point, isolating shape effects from amplitude effects.

- Families compared: symmetric logistic, hyperbolic-tangent, piecewise-linear breakpoint.
- Calibration constraint: matched asymptote and response level at $t = 5$ weeks via uniroot or closed form.
- Families differ in smoothness, inflection symmetry, and curvature while sharing the same marginal effect size.

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Three alternative families serve as sensitivity-analysis comparators, and in this section we shall describe each in turn before the simulation study proper. Each is parameterized so that the asymptotic maximum and a single-anchor calibration point (the response level at $t = 5$ weeks, calibrated to the Gompertz family at the same anchor) are matched across families. That is, the families differ in shape envelope (smoothness, symmetry around the inflection, and curvature) while sharing the same marginal effect size at the anchor, a design choice that we discuss further in section 4.

4.1 Symmetric logistic

The logistic function

$$y(t) = \frac{\text{maxr}}{1 + \exp(-\text{rate} \cdot (t - t_0))} - \text{offset} \quad (2)$$

[19]

The symmetric logistic rises faster than Gompertz before the inflection and slower after, with symmetric inflection as the principal structural contrast.

- Symmetric around inflection at t_0 ; a vertical offset ensures $y(0) = 0$.
- At matched asymptote and anchor, the logistic envelope differs visibly from Gompertz in pre- and post-inflection curvature.
- Inflection symmetry is the primary qualitative contrast with the asymmetric Gompertz.

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is monotone-saturating and symmetric around its inflection at $t = t_0$. A vertical offset is subtracted so that $y(0) = 0$. At matched maxr and matched response level at the calibration anchor, the logistic family rises faster than Gompertz before the inflection and slower after, producing a visibly different shape envelope while sharing the same asymptote. The symmetric inflection is the principal qualitative contrast with the asymmetric Gompertz curve, and the reader will note that this symmetry property is, in a sense, the logistic family's most theoretically interesting feature for the purposes of the present comparison.

4.2 Hyperbolic-tangent

The hyperbolic-tangent variant

$$y(t) = \text{maxr} \cdot \frac{\tanh(\text{rate} \cdot (t - t_0)) + 1}{2} - \text{offset} \quad (3)$$

[20]

The hyperbolic-tangent family represents the fast- saturating smooth regime that neither the logistic nor Gompertz can span with standard parameterizations.

- Symmetric and monotone-saturating, but with a steeper pre-asymptote rise than the logistic.
- Used in pharmacokinetic modeling for drugs with rapid receptor saturation.

- Its inclusion spans the fast-saturating smooth regime absent from the other three families.

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is also monotone-saturating and symmetric, but with a qualitatively faster approach to the asymptote than the logistic. The tanh family produces the steepest pre-asymptote rise of the three smooth families, and it has been used in pharmacokinetic modeling for drugs with rapid receptor saturation. We include it here as the representative of the fast-saturating smooth regime that neither the logistic nor the Gompertz family can span with their standard parameterizations.

4.3 Piecewise-linear breakpoint

The piecewise-linear-breakpoint family, formalized in the broken-line regression literature²⁵,

$$y(t) = \text{maxr} \cdot \min(1, t/t_{\text{bp}}) + \text{post_slope} \cdot \max(0, t - t_{\text{bp}}) \quad (4)$$

[21]

The piecewise-linear breakpoint is the canonical non- smooth alternative and provides the strongest test of the framework’s sensitivity to trajectory non-smoothness.

- A linear ramp to breakpoint t_{bp} then constant or small post-breakpoint slope.
- The breakpoint is the sole non-differentiable feature across the four families.
- Whether the Wald test detects the non-smooth feature is the central empirical question the simulation addresses.

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is the canonical non-smooth alternative: a linear ramp up to a breakpoint t_{bp} , then constant (when $\text{post_slope} = 0$) or with a small post-breakpoint slope. The breakpoint is the only non-differentiable feature across the four families and is the principal qualitative contrast with the three smooth families. This family is the strongest available test of the framework’s sensitivity to non-smoothness in the trajectory. But how different is the piecewise-linear breakpoint from the smooth families, from the perspective of the Wald test that the analysis model applies? That question is precisely what the simulation in section 4 is designed to answer.

5 Sensitivity-analysis simulation design

538

[22]

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The primary estimand is the `bm:Dbc` interaction coefficient; performance measures are power, Type I error, bias, and empirical MCSE.

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- Analysis model: $Sx \sim bm + t + Dbc + bm:Dbc + (1 \mid ptID)$ with `corCAR1(form = ~t|ptID)`.
- Primary outputs: power at $\alpha = 0.05$ and Type I error under $\beta_{bm:D} = 0$.
- Secondary outputs: bias and empirical MCSE of $\hat{\beta}_{bm:D}$; all proportions reported with binomial MCSE.

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The primary estimand is the biomarker-treatment interaction coefficient $\hat{\beta}_{bm:D}$ from the linear mixed-effects analysis model $Sx \sim bm + t + Dbc + bm:Dbc + (1 \mid ptID)$ with `corCAR1(form = ~t|ptID)`. Primary inferential outputs are power for $H_0 : \beta_{bm:D} = 0$ at $\alpha = 0.05$, and Type I error under $\beta_{bm:D} = 0$. Secondary estimands are bias of $\hat{\beta}_{bm:D}$ relative to the calibrated true effect and the empirical MCSE of $\hat{\beta}_{bm:D}$ across replicates. Performance measures are reported with binomial-proportion MCSEs for proportions and the standard MCSE of the within-replicate mean for continuous quantities.

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[23]

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The ADEMP design is pre-registered and fixes all four DGP families, the calibration sub-study, and the full factorial before production runs.

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- Pre-registration location: `analysis/scripts/gompertz-evaluation/00-ademp-pre-registration`
- Study 0 calibrates all four families to a common marginal effect size; Study 1 is the 144-cell factorial; Study 2 is the sloppiness eigenvalue diagnostic.
- Pre-registration prevents post-hoc specification of performance measures or DGP choices.

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The full Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures are pre-registered at `analysis/scripts/gompertz-evaluation/00-ademp-pr` which fixes the four DGP families (modified Gompertz, symmetric logistic, hyperbolic-tangent, piecewise-linear breakpoint), the calibration sub-study (Study 0) that matches all four to a common marginal effect size, the four-family power-and-bias factorial (Study 1, 144 cells), and the sloppiness eigenvalue diagnostic (Study 2) before the production runs.

[24]

The present submission uses a focused 16-cell factorial scoped to answer whether trajectory family matters for the bm:Dbc Wald test at MCSE 0.013–0.016.

- Design: 4 families $\times$ 2 architectures $\times$ 2 effect sizes; prazosin-calibrated OL+BDC, $N = 35$, $t_{1/2} = 0$.
- Carryover is absent to isolate trajectory shape from carryover-mediated correlation decay.
- The 144-cell extended grid is deferred to a follow-up; the 16-cell factorial addresses the central question at adequate MCSE resolution.

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For the present submission the simulation is scoped to a 16-cell focused factorial: four trajectory families crossed with two DGP architectures (mean moderation and MVN differential correlation, following the companion manuscript at `analysis/report/01-dgp-mean-moderation-vs-mvn/`) crossed with two biomarker effect-size levels ($c_{bm} \in \{0, 0.45\}$). The trial design is the prazosin-calibrated Hybrid OL+BDC, $N = 35$, $t_{1/2} = 0$, with carryover set to absent in order to isolate the trajectory-shape effect from carryover-mediated correlation decay. Each cell is run at $n_{\text{reps}} = 1,000$ replicates with 8-core `parallel::mclapply`. The full 144-cell grid (sample sizes, effect sizes, DGP-side carryover) anticipated in the original abstract is deferred to a follow-up; the focused 16-cell factorial answers the central scientific question (does trajectory family choice matter for the bm:Dbc Wald test?) at MCSE 0.013–0.016 per cell.

Notation for the moderation parameter. Because this factorial crosses both architectures, c_{bm} is used here as a common label for the biomarker-moderation strength at a matched numeric value. Strictly, the parameter is the correlation c_{bm} under Architecture B and the dimensionless multiplier β_{bm} under Architecture A, the two being calibrated to a common scale because the Architecture-A shift is applied as $\beta_{bm} b_i \sigma_{BR}$ on the standardized biomarker b_i . The companion architecture manuscript keeps the symbols distinct; we adopt the shared label only where a statement applies to both architectures at the matched value, and the architecture is named explicitly wherever a result is architecture-specific.

[25]

All four families are matched at $t = 5$ weeks with the asymptotic maximum held constant; calibrated parameters are recorded for reproducibility.

- Anchor calibration: each family's response at $t = 5$ matched to Gompertz via uniroot or closed form.
- Asymptotic maximum held constant across all four families.
- Calibrated parameters recorded in 02-faithful-trajectory-sweep.R and transcribed into the summary at 07-gompertz-faithful-summary.txt.

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Calibration. The four trajectory families are matched at a single anchor point ($t = 5$ weeks): each family's response level at $t = 5$ is set equal to the Gompertz reference level at $t = 5$ via uniroot (logistic, hyperbolic-tangent) or closed form (piecewise-linear breakpoint). The asymptotic maximum is held constant across families. Calibrated parameters are recorded in the driver analysis/scripts/gompertz-evaluation/02-faithful-trajectory-sweep.R.

[26]

The trajectory family drives the data-generating process directly, and under Architecture A the biomarker moderation is trajectory-scaled, in keeping with the paper-01 multiplicative model.

- The family sets the BR mean curve inside the covariance construction (buildSigma br_family), not by a post-hoc additive shift.
- Under Architecture A the per-timepoint moderation scales with the response trajectory, normalised so the mean on-drug magnitude equals σ_{BR} (matched marginal effect).
- Under Architecture B the interaction is in the covariance and is mean-independent, so the family enters the BR mean only.

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Trajectory-family encoding. The family is routed through the data-generating process rather than imposed post hoc: it sets the BR mean curve inside the covariance construction via the br_family argument to buildSigma. Under Architecture A (direct mean moderation) the biomarker-treatment interaction is implemented in the multiplicative form of the companion manuscript at analysis/report/01-dgp-mean-moderation-vs-mvn/, $Y_{it} = \mu_0 + \beta_1 D_{it}(1 + \beta_{bm} B_i) + \epsilon_{it}$, so the biomarker-dependent component of the treatment effect rides the response trajectory $BR(t)$ rather than entering as a flat shift. We normalise the per-timepoint moderation so that its mean on-drug magnitude equals σ_{BR} , holding the marginal interaction effect size fixed across families while letting the temporal profile vary by family shape. Under Architecture B (MVN differential

correlation) the interaction is encoded in the BM-BR covariance block, which is independent of the BR mean; the family therefore enters only through the mean curve and, as we verify by a matched-seed check, cannot move the interaction test.

6 Results

[27]

The 16-cell simulation completed with 100% convergence in 597 s; cell MCSE on power runs from 0.013 to 0.014.

- 1,000 replicates per cell across 16 cells; 8-core `parallel::mclapply`.
- Per-replicate output at `analysis/data/quick-sim/07-gompertz-faithful-replicates.rds`; ADEMP summary at `07-gompertz-faithful-summary.txt`.
- Results are presented in order: power, Type I error, bias.

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The simulation program was executed at 1{,}000 replicates per cell across the 16-cell trajectory-family-by-architecture-by- effect-size factorial in 597 s using 8-core `parallel::mclapply`. Convergence was 100% across all 16{,}000 fits. Per-replicate output is at `analysis/data/quick-sim/07-gompertz-faithful-replicates.rds`; the Morris ADEMP cell-level summary is at `analysis/data/quick-sim/07-gompertz-faithful-summary.txt`. Cell MCSE on power runs from 0.013 to 0.014 across the alternative cells and from 0.004 to 0.006 on the conservative null cells. We begin the results section with power, then turn to Type I error, and finally consider bias of the interaction estimate.

6.1 Power across response-curve families

Trajectory-family power at $c_{bm} = 0.45$, percent $\pm$ MCSE. Architecture A is the faithful trajectory-scaled mean-moderation form; Architecture B is the MVN covariance channel:

Family	Architecture A	Architecture B
Gompertz	74.3 $\pm$ 1.4	77.4 $\pm$ 1.3
Logistic	78.2 $\pm$ 1.3	79.1 $\pm$ 1.3
Hyperbolic tangent	76.0 $\pm$ 1.4	77.7 $\pm$ 1.3
Piecewise-linear breakpoint	77.8 $\pm$ 1.3	80.4 $\pm$ 1.3

[28]

The inferential consequence of trajectory family is architecture-dependent: exactly zero under the covariance channel, modest but non-zero under faithful mean moderation.

- Architecture B: with matched seeds the four families return identical rejection decisions, so family is inert by construction; the production-seed spread is sampling noise.
- Architecture A: families span 0.743 to 0.782, a 0.039 spread; Gompertz (the default) is the lowest-power family, roughly two MCSE below logistic.
- The pre-registered hypothesis that the piecewise-linear breakpoint would behave qualitatively differently is not supported; it tracks the logistic, not the Gompertz.

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The two architectures behave qualitatively differently. Under Architecture B (MVN differential correlation), the biomarker-treatment interaction is carried entirely by the covariance structure, which is independent of the BR mean curve; consequently the trajectory family, which sets that mean curve, cannot alter the interaction test. We confirm this directly: in a matched-seed check the four families return bit-for-bit identical rejection decisions, so the production-seed cell spread (0.774 to 0.804) is sampling noise rather than a family effect. Under Architecture A (direct mean moderation), implemented in the faithful multiplicative form in which the biomarker-dependent component of the treatment effect rides the response trajectory, the families separate: power ranges from 0.743 (Gompertz) through 0.760 (hyperbolic-tangent) and 0.778 (piecewise-linear) to 0.782 (logistic), a 0.039 spread. The Gompertz-logistic gap of 0.039 is approximately two standard errors of the difference ($SE \approx 0.019$), and the Gompertz default is the lowest-power, that is the mildly conservative, family of the four. The pre-registered hypothesis that the piecewise-linear breakpoint family would behave qualitatively differently from the smooth families is not supported in the direction anticipated: the breakpoint family tracks the logistic and the hyperbolic-tangent, not the Gompertz default.

[29]

The earlier flat-moderation result, which reported no family effect under either architecture, was an artefact of the interaction encoding, not a property of trajectory family.

- The prototype applied the biomarker moderation as a flat shift, independent of the response trajectory, so family could not affect the interaction under either architecture.
- Once the moderation is trajectory-scaled per the paper-01 Architecture-A model, the mean-moderation families separate.
- The covariance-channel invariance is unchanged, because that channel is genuinely mean-independent.

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An earlier draft of this manuscript reported essentially no family effect under either architecture. That result is now understood to have been an artefact of the interaction encoding rather than a property of trajectory family. In that draft the Architecture-A biomarker moderation was applied as a flat per-timepoint shift of magnitude $\beta_{bm}\sigma_{BR}$, independent of the response trajectory, so that the trajectory family, which enters only through the BR mean curve, could not affect the bm:Dbc interaction under Architecture A any more than under the mean-independent Architecture B. Implementing Architecture A in its faithful multiplicative form (section 4), in which the biomarker-dependent component scales with the response trajectory in keeping with the companion manuscript at [analysis/report/01-dgp-mean-moderation-vs-mvn/](#), the mean-moderation families separate at the few-percent level while the covariance-channel invariance under Architecture B is preserved, the latter being a genuine mean-independence property rather than an artefact.

6.2 Type I error across response-curve families

Type I error at $c_{bm} = 0$, percent $\pm$ MCSE:

Family	Architecture A	Architecture B
Gompertz	1.9 ± 0.4	2.6 ± 0.5
Logistic	3.8 ± 0.6	2.2 ± 0.5
Hyperbolic tangent	2.7 ± 0.5	2.6 ± 0.5
Piecewise-linear breakpoint	2.8 ± 0.5	2.7 ± 0.5

[30]

Type I error is slightly conservative across all four families, consistent with corCAR1 small-sample degree-of-freedom accounting.

- All eight null cells fall in $[0.019, 0.038]$; no family-specific inflation is observed.
- At the null the moderation term vanishes, so the trajectory-scaling of Architecture A is inactive and the null cells are governed by the same mechanism as before.
- The conservatism is shared across families, implicating the corCAR1 d.f. mechanism rather than trajectory shape.

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All eight null cells produce empirical Type I error in $[0.019, 0.038]$, slightly conservative relative to the nominal 0.05 and consistent with the corCAR1 small-sample degree-of-freedom accounting at the prazosin-calibrated $N = 35$ and 8-timepoint short serial series. At the null the biomarker moderation term is zero, so the trajectory-scaling that distinguishes the Architecture-A families under the alternative is inactive here, and the null cells differ across families only by sampling noise. There is no family-specific inflation; the highest cell, logistic under Architecture A at 0.038, is within two MCSE of the others. The conservatism is shared across all four families, supporting the interpretation that it reflects the corCAR1 d.f. mechanism rather than a trajectory-shape artefact, and it has appeared consistently across pmsimstats simulation runs at the reference parameters.

6.3 Bias of the biomarker-treatment interaction estimate

[31]

Bias of the interaction estimate is negligible and family-invariant under both architectures, so the Architecture-A power separation is a precision effect, not a bias effect.

- Mean $\hat{\beta}_{bm:D_{bc}}$ varies from -0.230 to -0.239 across the eight alternative cells, a 0.009 spread within Monte Carlo noise.
- Empirical SE is essentially constant across families (0.073 to 0.083).
- The matched-anchor calibration holds the estimand fixed across families, so the power spread reflects the temporal profile of the signal rather than its magnitude or bias.

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Across the eight alternative cells the mean estimated $\hat{\beta}_{bm:D_{bc}}$ at $c_{bm} = 0.45$ varies over a narrow range (roughly -0.230 to -0.239), a cell-to-cell spread of about 0.009 that is itself within the Monte Carlo noise of the estimator and shows no family-specific drift; the Gompertz and logistic cells that differ most in power have nearly identical mean estimates. The empirical SE of $\hat{\beta}_{bm:D_{bc}}$ is essentially constant across families (SD in the range 0.073 to 0.083 across cells) and roughly equal between architectures. That the Architecture-A power separation appears with negligible bias and near-constant empirical SE indicates it is a precision effect operating through the test statistic's sensitivity to the temporal profile of the signal, not a shift in the estimand or its sampling dispersion. The matched-anchor calibration holds the marginal effect size fixed across families by construction, so the residual family differences are differences in the temporal shape of the moderation, which is precisely the axis the design isolates.

7 Discussion

7.1 An architecture-dependent finding

[32]

The inferential consequence of trajectory family depends on how the biomarker-treatment interaction is encoded: zero under the covariance channel, modest under mean moderation.

- Under Architecture B (covariance channel) the family is inert by construction; the interaction test is mean-independent.
- Under Architecture A (mean moderation) the four families separate over a 0.039 power range, with the Gompertz default the lowest-power, that is mildly conservative, family.
- The effect is modest (about two MCSE at $N = 35$, $c_{bm} = 0.45$) and should not be overstated, but it is real and direction-consistent.

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The central finding is that the inferential consequence of trajectory-family choice is architecture-dependent. At the prazosin-calibrated reference cell (Hybrid OL+BDC, $N = 35$, $c_{bm} = 0.45$, $t_{1/2} = 0$, matched-anchor calibration), the four families span the qualitative range of plausible templates for monotone-saturating treatment-response trajectories: asymmetric (Gompertz), symmetric smooth (logistic and hyperbolic-tangent), and non-smooth (piecewise-linear breakpoint). Under Architecture B, where the interaction is carried by the BM-BR covariance and is therefore independent of the BR mean curve, the family is inert by construction. Under Architecture A, where the interaction is a mean effect that, in its faithful multiplicative form, scales with the response trajectory, the families separate over a 0.039 power range and the Gompertz default is the lowest-power, that is the mildly conservative, family. We would not wish to overstate the Architecture-A effect; it is modest, on the order of two Monte Carlo standard errors at this sample size and effect size, and the conditions under which it would grow or shrink are discussed in section 6.3. But it is real and direction-consistent, and it overturns the apparent null reported in the earlier flat- moderation draft.

[33]

The pre-registered piecewise-linear hypothesis is not supported in the anticipated direction; the breakpoint family tracks the smooth alternatives, not the Gompertz default.

- The non-differentiable breakpoint carries no distinctive power sig-

- 856 nature at $N = 35$ and $c_{bm} = 0.45$.
- 857 • Under Architecture A the breakpoint family sits with the logistic
- 858 and hyperbolic-tangent, above the Gompertz default.
- 859 • Smoothness per se is not the operative axis; the temporal profile
- 860 of the moderation is.

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863 The pre-registered hypothesis that the piecewise-linear breakpoint
864 family, with its non-differentiable feature, would behave qualitatively
865 differently from the smooth families is not supported in the direction
866 anticipated. The breakpoint family carries no distinctive power
867 signature attributable to its non-smoothness; under Architecture A
868 it sits with the logistic and hyperbolic-tangent families, above the
869 Gompertz default, and under Architecture B it is inert like the others.
870 The operative axis is therefore not smoothness but the temporal
871 profile of the biomarker-moderated signal: families whose response
872 rises more gradually across the on-drug timepoints, and so carry more
873 of the moderated signal at the later, more informative timepoints,
874 yield marginally higher power than the front-loaded Gompertz curve,
875 which reaches near-asymptote early.

876 7.2 Why the architectures differ

877 [34]

878 **Two mechanisms explain the contrast: exact mean- indepen-**
879 **dence of the covariance channel, and the residual profile-**
880 **sensitivity of the trajectory-scaled mean channel.**

- 881 • Under Architecture B the biomarker-treatment information lives
- 882 in the covariance and is mathematically independent of the tra-
- 883 jectory mean, so the family cannot move the test.
- 884 • Under Architecture A the moderation rides the trajectory, so the
- 885 temporal profile of the signal, not just its anchored magnitude,
- 886 enters the Wald statistic.
- 887 • Matched-anchor calibration leaves only profile differences, which
- 888 the time-averaging Wald test damps but does not erase.

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891 We see two mechanisms that, taken together, account for the architec-
892 ture contrast, and we shall work through each in turn. The first is the
893 exact mean-independence of the covariance channel under Architec-
894 ture B; the second is the residual profile-sensitivity of the trajectory-
895 scaled mean channel under Architecture A, which the matched-anchor
896 calibration and the time-averaging of the Wald test damp but do not

897 eliminate.

898 [35]

899 **Matched-anchor calibration ensures that residual family dif-**
900 **ferences are in shape only, not in amplitude, and those shape**
901 **differences are small relative to between-replicate variation.**

- 902 • Families matched at asymptote and $t = 5$ anchor.
- 903 • Within-trajectory shape variation is small relative to the between-
- 904 replicate variance of the bm:Dbc estimator.
- 905 • The calibration design isolates shape effects from amplitude ef-
- 906 fects by construction.

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909 The first mechanism damps the Architecture-A effect: calibration re-
910 moves the amplitude differences, leaving only profile differences. The
911 four families are matched at the asymptotic maximum and at a sin-
912 gle anchor point ($t = 5$ weeks), and the trajectory-scaled moderation
913 is normalised to a common mean on-drug magnitude. At matched
914 marginal effect size, the residual within-trajectory variation in shape
915 is small relative to the between-replicate variation in the bm:Dbc es-
916 timator. The calibration constraint is intentional in the simulation
917 design: it isolates trajectory-shape effects from amplitude effects, so
918 that the family-specific power differences we observe under Architec-
919 ture A can be attributed to the temporal profile of the moderation
920 alone, and explains why that effect is on the order of a few percentage
921 points rather than larger.

922 [36]

923 **The bm:Dbc Wald test averages over on-drug timepoints,**
924 **smoothing out local trajectory-shape differences that would**
925 **be visible in point-in-time comparisons.**

- 926 • The continuous D_{bc} predictor integrates trajectory contributions
- 927 across all on-drug timepoints.
- 928 • Local shape differences are averaged within the integration and
- 929 do not accumulate to detectable power differences.
- 930 • This temporal averaging damps, but does not erase, the profile
- 931 differences that drive the Architecture-A spread.

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934 The second mechanism, also damping, is that the bm:Dbc Wald test
935 averages over time. The continuous D_{bc} predictor multiplied by the
936 biomarker integrates the trajectory's contribution across all on-drug

timepoints, with weights determined by the predictor’s exponential-decay structure. Trajectory-shape differences (where one family rises faster than another between timepoints) are partly averaged within the integration, so the Wald test is relatively, though not perfectly, insensitive to profile features. Under Architecture A the averaging is incomplete: the residual profile difference survives into the test statistic and produces the observed few-percent power spread. Under Architecture B there is no profile term to survive, because the interaction does not enter the mean at all.

[37]

Under Architecture B the interaction is in the covariance structure and is exactly independent of trajectory shape, which we verify by a matched-seed check.

- Architecture B encodes information in $\partial\text{Cor}(BM, BR)/\partial D_{bc}$, not in the trajectory mean.
- Holding the seed fixed, the four families return identical rejection decisions, so the invariance is exact, not merely within Monte Carlo noise.
- This is the mechanism behind the exact Architecture-B inertness, in contrast to the merely damped Architecture-A effect.

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Mean-independence applies specifically to Architecture B: the covariance moderation is upstream of trajectory shape. Under MVN differential correlation, the biomarker-treatment information lives in the $\partial\text{Cor}(BM, BR)/\partial D_{bc}$ structure, which is independent of the trajectory family that BR follows; the family enters only the BR mean, to which the interaction test is invariant. This is not a small-effect statement but an exact one, and we verify it directly: holding the random-number seed fixed across families, the four Architecture-B cells return bit-for-bit identical rejection decisions. The production-seed spread under Architecture B is therefore sampling noise, whereas the Architecture-A spread is a genuine, if modest, profile effect.

[38]

The two mechanisms together place the modified Gompertz default as inferentially free under the covariance channel and mildly conservative under the mean channel.

- Under Architecture B the default is robust by exact construction; trajectory family cannot matter.
- Under Architecture A the default is the lowest-power family, so it is a safe, slightly conservative, choice rather than an optimal

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- one.
- The following section identifies conditions that would amplify the Architecture-A effect.

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The combined effect is that trajectory-family choice is inferentially free under the covariance channel and carries a modest, mildly conservative, cost under the mean channel. The framework’s modified Gompertz default is robust by exact construction under Architecture B, and under Architecture A it is the lowest-power of the four families, which makes it a safe rather than an optimal default: a trial designer who adopts it under a mean-moderation DGP will, if anything, slightly under-state achievable power. We turn next to the conditions under which the Architecture-A effect would be expected to grow.

7.3 Conditions that would change the answer

[39]

Four conditions outside the present 16-cell factorial would plausibly amplify the Architecture-A family effect; none would activate an effect under the mean-independent Architecture B.

- Smaller N , different c_{bm} , non-zero carryover ($t_{1/2} > 0$), and non-monotone or biphasic shape envelopes.
- None of these conditions is exercised in the present focused factorial.
- Each is a natural axis for the 144-cell follow-up grid; the biphasic case is the only one expected to reach Architecture B.

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Four conditions, none exercised in the present 16-cell factorial, would plausibly amplify the modest Architecture-A family effect, and each merits a brief comment. Because Architecture B is mean-independent, the first three conditions (sample size, effect size, carryover) bear on Architecture A only; the fourth, non-monotone shape, is the only one that could in principle reach Architecture B, and then only by changing the covariance construction rather than the mean.

[40]

Smaller sample size might amplify local-shape sensitivity by concentrating temporal information at fewer timepoints.

- At $N = 35$, the test integrates over a moderately large within-subject sample.

- With fewer timepoints, local shape differences between families would be more visible rather than averaged.
- This condition is the most proximate candidate for amplifying the Architecture-A effect beyond a few percent.

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First, smaller sample size. At $N = 35$ the bm:Dbc test integrates over a moderately large within-subject sample; smaller N might amplify the local-shape sensitivity by concentrating the temporal information at fewer timepoints where the families differ more visibly.

[41]

At smaller c_{bm} near the 50% power boundary, family differences might surface via different finite-sample sampling distributions.

- The $c_{bm} = 0.45$ cell sits in a moderate-power regime with comparable signal-to-noise across families.
- Pushing to $c_{bm} = 0.30$ (power near 0.5) may reveal family-specific finite-sample sampling differences.
- The tight clustering at 0.45 may be a local feature of the effect-size space.

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Second, larger or smaller c_{bm} . The $c_{bm} = 0.45$ reference cell sits in a power regime where families have comparable signal-to-noise. Pushing c_{bm} to 0.30 (where power lies near 0.5) might surface family differences via different finite-sample sampling distributions, and it is possible that the tight clustering we observe is a feature of this particular region of the effect-size space.

[42]

Non-zero carryover introduces an interaction between trajectory shape and the off-drug exponential decay that is suppressed in the present $t_{1/2} = 0$ slice.

- With $t_{1/2} > 0$, integration weights interact with the off-drug decay, potentially differentiating families.
- The present simulation sets $t_{1/2} = 0$ to isolate trajectory shape from carryover effects.
- This interaction is the mechanistic pathway by which carryover could make trajectory family consequential.

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Third, carryover-induced correlation decay. With $t_{1/2} > 0$, the bm:Dbc-mediated information channel is eroded differently for different trajectory shapes, because the integration weights interact with the off-drug exponential decay. The $t_{1/2} = 0$ slice exercised here suppresses this interaction entirely.

[43]

Non-monotone or biphasic trajectory families would produce genuinely different bm:Dbc information content to which the Wald test would almost certainly respond.

- All four present families share a monotone-saturating envelope; the small-effect finding is conditional on this property.
- Biphasic or oscillating families have qualitatively different information structure.
- Extension to non-monotone families is outside the scope of the present study.

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Fourth, non-monotone or biphasic shape envelopes. The four families we have examined share a monotone-saturating envelope. Families that peak and decay (biphasic) or oscillate would produce genuinely different trajectories with different bm:Dbc information content, and the Wald test would almost certainly respond to those differences.

[44]

The 144-cell extended grid is the canonical follow-up that would close the trajectory-family question at publication-adequate rigour.

- The grid crosses N , c_{bm} , $t_{1/2}$, and family.
- Each of the four conditions above maps to a dimension of this grid.
- The present focused factorial is a necessary precursor; the extended grid is the production-grade completion.

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Each condition is the natural axis of a follow-up sensitivity analysis. The 144-cell grid anticipated in the abstract ($\$N \times c_{\{bm\}} \times t_{\{1/2\}} \times \$family$) is the canonical follow-up, and we recommend it as the production-grade extension that would close out the trajectory-family question at a level of rigour adequate for the methodological literature.

7.4 The prazosin/PTSD reference cell

[45]

For the prazosin-PTSD application, trajectory family is a low-consequence sensitivity axis relative to carryover and DGP-architecture, but not a free one under mean moderation.

- Sensitivity to the modified Gompertz function is small relative to carryover specification and DGP-architecture sensitivity.
- Under a covariance-channel DGP the modified Gompertz choice is inferentially free; under mean moderation it is mildly conservative, costing a few percentage points of power.
- Sensitivity-analysis resources are better spent on carryover and DGP-architecture axes with larger inferential consequences, while still reporting one family comparison under the mean-moderation DGP.

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For the motivating prazosin-PTSD application that drives the pmsimstats research program, the present result implies that sensitivity to the modified Gompertz assumption is small relative to sensitivity to the carryover specification (the subject of the companion manuscript at `analysis/report/02-carryover-sensitivity/`) and to the DGP architecture (the subject of `analysis/report/01-dgp-mean-moderation-vs-mvn/`). Of the three principal axes of sensitivity uncertainty in the pmsimstats simulation framework, trajectory family choice is, at the reference parameters, the least consequential by a substantial margin. It is not, however, entirely free: under the mean-moderation DGP the modified Gompertz default is the lowest-power family by a few percentage points, so a trial designer who adopts it is making a mildly conservative choice. Under the covariance-channel DGP the choice is inferentially free. We would suggest that sensitivity-analysis attention be directed chiefly to the carryover and DGP-architecture axes, where the inferential consequences are larger, while reporting at least one Gompertz-vs-alternative comparison whenever the operative DGP is mean moderation.

8 Conclusions

[46]

The modified Gompertz is a defensible template for monotone-saturating N-of-1 trajectories; the inferential cost of the choice is architecture-dependent.

- Under the covariance channel (Architecture B) the four families

- are exactly equivalent; under mean moderation (Architecture A) they span a 0.039 power range at MCSE 0.013–0.014.
- The Gompertz default is the lowest-power family under mean moderation, that is mildly conservative, not optimal.
 - The piecewise-linear breakpoint hypothesis is not supported; the breakpoint family tracks the smooth alternatives.

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First, the modified Gompertz function is a defensible parametric template for monotone-saturating treatment-response trajectories in N-of-1 simulation, with an architecture-dependent qualification. At the prazosin/PTSD reference cell with matched-anchor calibration, the four-family comparison (Gompertz, symmetric logistic, hyperbolic-tangent, piecewise-linear breakpoint) produces exactly equivalent inference under the covariance-channel architecture, where the interaction is mean-independent, and a modest 0.039 power spread under the mean-moderation architecture, where the biomarker moderation rides the response trajectory. In the latter case the Gompertz default is the lowest-power of the four families, which makes it a mildly conservative rather than an optimal default. The pre-registered hypothesis that the piecewise-linear breakpoint family would behave qualitatively differently from the smooth families is not supported in the direction anticipated.

[47]

The finding implies the Wald test is robust to trajectory family under the covariance channel and mildly sensitive to it under mean moderation; boundary conditions are deferred to the 144-cell follow-up.

- Robustness is exact under Architecture B and approximate (few-percent) under Architecture A in the evaluated regime.
- At what sample sizes, effect sizes, or carryover levels the Architecture-A effect grows is an open question.
- Sensitivity-analysis priority should go to carryover and DGP-architecture axes with larger demonstrated consequences.

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Second, the finding is methodologically informative precisely because it is architecture-dependent. It implies that the bm:Dbc Wald test is exactly robust to trajectory-family choice when the interaction is encoded in the covariance, and only mildly sensitive to it, at the level of a few percentage points of power, when the interaction is a trajectory-scaled mean effect. We cannot yet establish at what sample sizes,

effect sizes, or carryover levels the Architecture-A effect would grow; that question is deferred to the 144-cell follow-up. Trial designers using the pmsimstats framework can adopt the modified Gompertz default with confidence under a covariance-channel DGP, and with the understanding that it is mildly conservative under a mean-moderation DGP, while directing sensitivity-analysis attention to the carryover and DGP-architecture axes that have larger inferential consequences.

[48]

The 144-cell extended grid is the work needed to close the trajectory-family question at publication-adequate rigour.

- The present 16-cell factorial addresses the central question with statistical clarity.
- The extended grid would map boundary conditions at which trajectory family begins to matter.
- This follow-up is recommended as the production-grade completion of the trajectory-family evaluation.

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Third, the 144-cell extended grid (sample sizes, effect sizes, DGP-side carryover) anticipated in the original abstract is the natural follow-up, and we regard it as the work that would be needed to close the trajectory-family question at a level of rigour adequate for the methodological literature. The present focused 16-cell factorial addresses the central scientific question with statistical clarity; the extended grid would map the boundary conditions at which trajectory-family choice begins to matter.

9 Conflict of interest

The authors declare no conflicts of interest.

10 Funding

This work received no specific funding.

11 Data availability statement

[49]

Data and code are openly available in the pmsimstats-ng repository under versioned standard paths.

- Per-replicate Monte Carlo outputs and ADEMP cell-level summaries are at `analysis/data/`.

- Simulation drivers are at `analysis/scripts/`.
- Exact manuscript paths are listed in the Reproducibility section.

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The data and code that support the findings of this study are openly available in the `pmsimstats-ng` project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

12 Reproducibility

[50]

The manuscript is fully reproducible inside the `zzcollab` Docker container with the `renv.lock` package set.

- Container image hash in `Dockerfile`; package set in `renv.lock`.
- Principal packages: `nlme`, `parallel`, `data.table`, `furrr/purrr`, `rmarkdown/bookdown`.
- Rendered with R 4.4.x.

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This manuscript was rendered with R 4.4.x inside the project’s `zzcollab` Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the original-extended simulation collection), `furrr` and `purrr` (the tidyverse simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build.

[51]

Seeds, checkpoints, and driver paths are fully specified for exact replication of the simulation results.

- Master seed `set.seed(20260617)`; per-cell seed offsets give independent but reproducible replicate streams.
- Outputs checkpointed at `analysis/data/quick-sim/07-gompertz-faithful-replicates.rds`; ADEMP summaries at `07-gompertz-faithful-summary.txt`.
- Driver `02-faithful-trajectory-sweep.R`; pre-registered ADEMP plan at `analysis/scripts/gompertz-evaluation/00-ademp-pre-registration.md`.

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The simulation driver `analysis/scripts/gompertz-evaluation/02-faithful-trajectory-sweep.R` sets `set.seed(20260617)` and assigns each of the 16 cells a distinct seed offset, so replicate streams are independent across cells but reproducible. The trajectory family is routed through the data-generating process via the `br_family` argument to `buildSigma`, and the Architecture-A moderation is trajectory-scaled via the `moderation_scaling = "trajectory"` argument to `generateData` (both default to the original behavior, so all other pmsimstats runs are unchanged). Per-replicate simulation outputs are checkpointed at `analysis/data/quick-sim/07-gompertz-faithful-replicates.rds`; Morris ADEMP cell-level summaries are at the companion `07-gompertz-faithful-summary.txt`. The pre-registered ADEMP plan is at `analysis/scripts/gompertz-evaluation/00-ademp-pre-registration.md`, and deviations from it are recorded in `02-deviations-log.md`. The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/07-gompertz-evaluation/`.

13 References

1. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: Statistical methods and practical considerations. *Frontiers in Digital Health*. 2020;2:13. doi:10.3389/fdgth.2020.00013
2. Laird AK. Dynamics of tumor growth. *British Journal of Cancer*. 1964;13(3):490-502. doi:10.1038/bjc.1964.55
3. Norton L. A Gompertzian model of human breast cancer growth. *Cancer Research*. 1988;48(24 Pt 1):7067-7071.
4. Zwietering MH, Jongenburger I, Rombouts FM, Riet K van 't. Modeling of the bacterial growth curve. *Applied and Environmental Microbiology*. 1990;56(6):1875-1881. doi:10.1128/aem.56.6.1875-1881.1990
5. Richards FJ. A flexible growth function for empirical use. *Journal of Experimental Botany*. 1959;10(2):290-301.
6. Tjørve E, Tjørve KMC. A unified approach to the Richards-model family for use in growth analyses: Why we need only two model forms. *Journal of Theoretical Biology*. 2010;267(3):417-425. doi:10.1016/j.jtbi.2010.09.008
7. Tjørve KMC, Tjørve E. The use of Gompertz models in growth analyses, and new Gompertz-model approach: An addition to the Unified-Richards family. *PLOS ONE*. 2017;12(6):e0178691. doi:10.1371/journal.pone.0178691

- 1278 8. Lillie EO, Patay B, Diamant J, Issell B, Topol EJ, Schork NJ. The n-of-1 clinical trial: The ultimate strategy for individualizing medicine? *Personalized Medicine*. 2011;8(2):161-173. doi:10.2217/pme.11.7
- 1279 9. Schork NJ, Beaulieu-Jones B, Liang WS, Smalley S, Goetz LH. Exploring human biology with N-of-1 clinical trials. *Cambridge Prisms: Precision Medicine*. 2023;1:e12. doi:10.1017/pcm.2022.15
- 1280 10. Vohra S, Shamseer L, Sampson M, et al. CONSORT extension for reporting N-of-1 trials (CENT) 2015 statement. *Journal of Clinical Epidemiology*. 2015;76:9-17. doi:10.1016/j.jclinepi.2015.05.004
- 1281 11. Benzekry S, Lamont C, Beheshti A, et al. Classical mathematical models for description and prediction of experimental tumor growth. *PLOS Computational Biology*. 2014;10(8):e1003800. doi:10.1371/journal.pcbi.1003800
- 1282 12. Goutelle S, Maurin M, Rougier F, et al. The Hill equation: A review of its capabilities in pharmacological modelling. *Fundamental and Clinical Pharmacology*. 2008;22(6):633-648. doi:10.1111/j.1472-8206.2008.00633.x
- 1283 13. Dayneka NL, Garg V, Jusko WJ. Comparison of four basic models of indirect pharmacodynamic responses. *Journal of Pharmacokinetics and Biopharmaceutics*. 1993;21(4):457-478. doi:10.1007/BF01061691
- 1284 14. Bertalanffy L von. Quantitative laws in metabolism and growth. *Quarterly Review of Biology*. 1957;32(3):217-231.
- 1285 15. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*. 2019;38(11):2074-2102.
- 1286 16. Jagadeesan P, Raman K, Tangirala AK. Sloppiness: Fundamental study, new formalism and its application in model assessment. *PLOS ONE*. 2023;18(3):e0282609. doi:10.1371/journal.pone.0282609
- 1287 17. Hill AV. The possible effects of the aggregation of the molecules of haemoglobin on its dissociation curves. *Journal of Physiology*. 1910;40:iv-vii.
- 1288 18. Raskind MA, Peterson K, Williams T, et al. A trial of prazosin for combat trauma PTSD with nightmares in active-duty soldiers returned from Iraq and Afghanistan. *American Journal of Psychiatry*. 2013;170(9):1003-1010. doi:10.1176/appi.ajp.2013.12081133

- 1289 19. Ebling WF, Matsumoto Y, Levy G. Feasibility of effect-controlled clinical trials of drugs with pharmacodynamic hysteresis using sparse data. *Pharmaceutical Research*. 1996;13(12):1804-1810. doi:10.1023/a:1016072806164
- 1290 20. Triggs EJ, Charles BG, Contin M, et al. Population pharmacokinetics and pharmacodynamics of oral levodopa in parkinsonian patients. *European Journal of Clinical Pharmacology*. 1996;51(1):59-67. doi:10.1007/s002280050161
- 1291 21. Hockenberry MJ, Hooke MC, Rodgers C, et al. Symptom trajectories in children receiving treatment for leukemia: A latent class growth analysis with multitrajectory modeling. *Journal of Pain and Symptom Management*. 2017;54(1):1-8. doi:10.1016/j.jpainsymman.2017.03.002
- 1292 22. Verhulst PF. Notice sur la loi que la population suit dans son accroissement. *Correspondance Mathématique et Physique*. 1838;10:113-121.
- 1293 23. Gompertz B. On the nature of the function expressive of the law of human mortality, and on a new mode of determining the value of life contingencies. *Philosophical Transactions of the Royal Society of London*. 1825;115:513-583.
- 1294 24. Winsor CP. The Gompertz curve as a growth curve. *Proceedings of the National Academy of Sciences USA*. 1932;18(1):1-8.
- 1295 25. Muggeo VMR. Estimating regression models with unknown break-points. *Statistics in Medicine*. 2003;22(19):3055-3071. doi:10.1002/sim.1545

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